

FULL STACK WEB DEVELOPMENT

Experiment No. 02

Design Webpage using HTML5 & CSS3

1. Basic Information

Experiment No.	02
Date	
Title	Design Webpage using HTML5 & CSS3

Course Outcome: CO1

2. Objective / Aim

To design and implement a simple webpage using HTML5 and CSS3, demonstrating basic and medium-level CSS concepts.

3. System Requirements

- Hardware: Processor – Core i5 or equivalent, RAM – 8 GB or above, Storage – 10 GB free space.
- Software: Windows 11 / Linux, Visual Studio Code, Google Chrome or any modern web browser.
- Technologies: HTML5 and CSS3.

4. Architecture / Design

The webpage uses HTML5 for the structure and CSS3 for presentation and styling.

HTML5 Structure → External CSS3 Styling → Web Browser → Styled Webpage

5. Source Code

File 1: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSS Demonstration</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>CSS Demonstration</h1>
    <p>Basic and Medium CSS Concepts</p>
  </header>

  <nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Courses</a>
    <a href="#">Contact</a>
  </nav>

  <main>
    <section>
      <h2>About CSS</h2>
      <p>
        CSS stands for Cascading Style Sheets.
        It is used to style and design HTML webpages.
      </p>
    </section>
  </main>
</body>
</html>
```

```

</section>

<section>
  <h2>My Skills</h2>
  <div class="cards">
    <div class="card">
      <h3>HTML</h3>
      <p>Used to create the structure of a webpage.</p>
    </div>

    <div class="card">
      <h3>CSS</h3>
      <p>Used to style and design the webpage.</p>
    </div>

    <div class="card">
      <h3>JavaScript</h3>
      <p>Used to add functionality and interaction.</p>
    </div>
  </div>
</section>

<section>
  <h2>Student Details</h2>
  <table>
    <tr>
      <th>Name</th>
      <th>Branch</th>
      <th>Year</th>
    </tr>
    <tr>
      <td>Venu Gopal</td>
      <td>Computer Engineering</td>
      <td>2nd Year</td>
    </tr>
    <tr>
      <td>Rahul</td>
      <td>Information Technology</td>
      <td>2nd Year</td>
    </tr>
  </table>
</section>

<section>
  <h2>Contact</h2>
  <form>
    <input type="text" placeholder="Enter your name">
    <input type="email" placeholder="Enter your email">
    <textarea placeholder="Enter your message"></textarea>
    <button type="submit">Submit</button>
  </form>
</section>
</main>

<footer>
  <p>© 2026 CSS Demonstration</p>
</footer>

</body>
</html>

```

File 2: style.css

```

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #f2f2f2;
  color: #333;
}

header {

```

```

        background-color: #2563eb;
        color: white;
        text-align: center;
        padding: 30px;
    }

    header h1 {
        margin: 0;
        font-size: 36px;
    }

    nav {
        background-color: #222;
        text-align: center;
        padding: 15px;
    }

    nav a {
        color: white;
        text-decoration: none;
        margin: 0 20px;
    }

    nav a:hover {
        color: #60a5fa;
    }

    main {
        width: 900px;
        margin: 30px auto;
    }

    section {
        background-color: white;
        margin-bottom: 25px;
        padding: 25px;
        border: 1px solid #ddd;
        border-radius: 10px;
    }

    h2 {
        color: #2563eb;
        border-bottom: 2px solid #2563eb;
        padding-bottom: 8px;
    }

    .cards {
        display: flex;
        gap: 20px;
    }

    .card {
        flex: 1;
        background-color: #eff6ff;
        padding: 20px;
        border-radius: 10px;
        box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
        transition: 0.3s;
    }

    .card:hover {
        transform: translateY(-5px);
        box-shadow: 0 8px 18px rgba(0, 0, 0, 0.2);
    }

    table {
        width: 100%;
        border-collapse: collapse;
    }

    th, td {
        border: 1px solid #ccc;
        padding: 12px;
        text-align: left;
    }

```

```

th {
  background-color: #2563eb;
  color: white;
}

tr:nth-child(even) {
  background-color: #f1f5f9;
}

input, textarea {
  width: 100%;
  padding: 12px;
  margin-bottom: 15px;
  border: 1px solid #ccc;
  border-radius: 6px;
}

input:focus, textarea:focus {
  border-color: #2563eb;
  outline: none;
}

button {
  background-color: #2563eb;
  color: white;
  padding: 12px 25px;
  border: none;
  border-radius: 6px;
  cursor: pointer;
  transition: 0.3s;
}

button:hover {
  background-color: #1d4ed8;
  transform: scale(1.05);
}

footer {
  background-color: #222;
  color: white;
  text-align: center;
  padding: 20px;
}

```

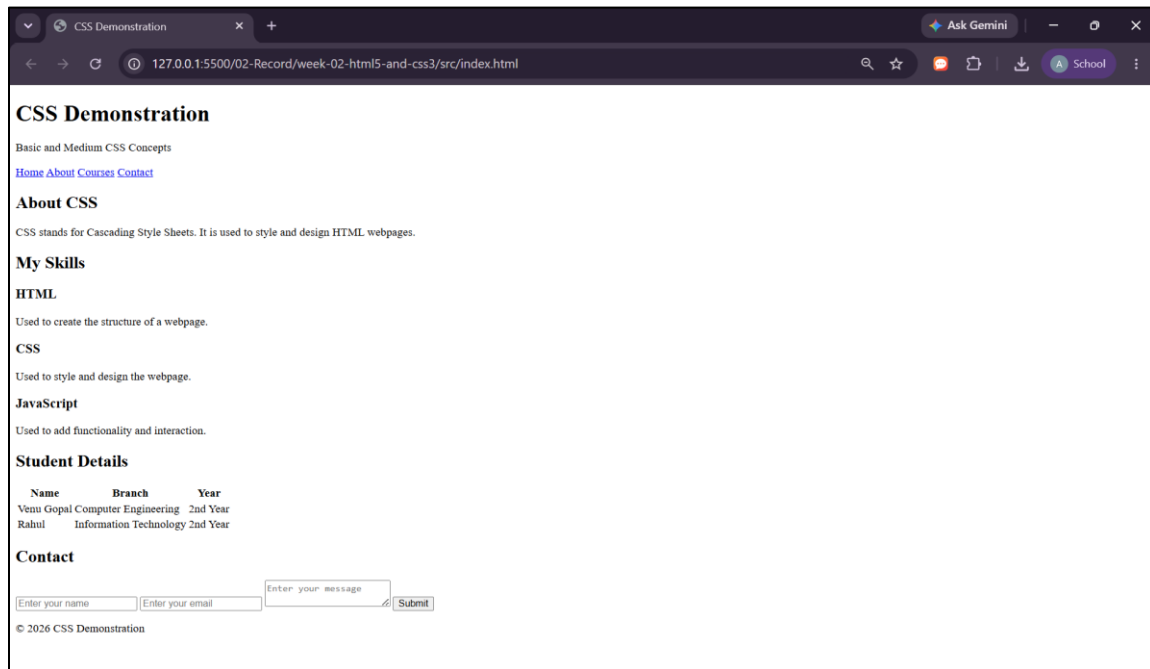
6. Compilation / Execution Steps

1. Create a folder and save the two files as index.html and style.css.
2. Make sure both files are present in the same folder.
3. Open index.html in Google Chrome or another web browser.
4. Observe the styled webpage and the CSS effects such as hover, focus, cards, table styling and transitions.

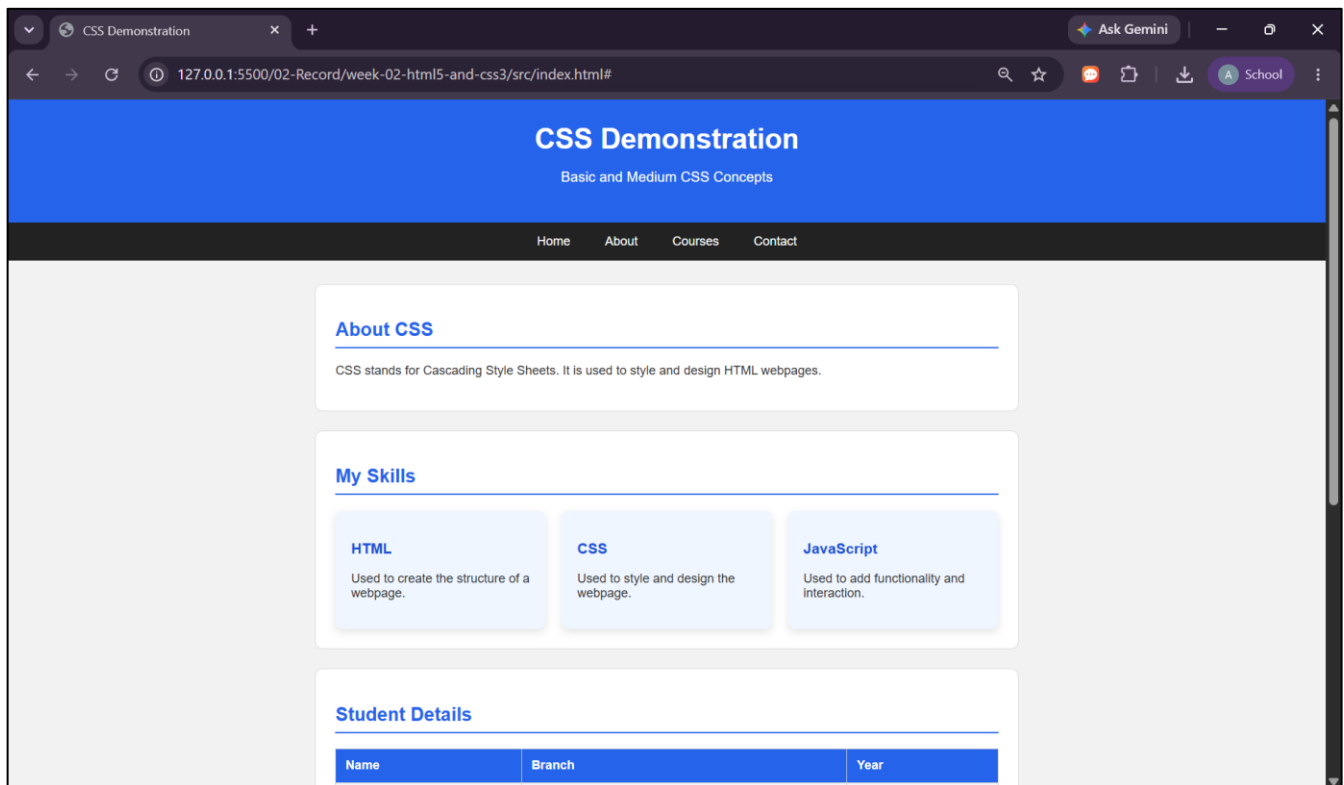
7. Expected Output / Screenshots

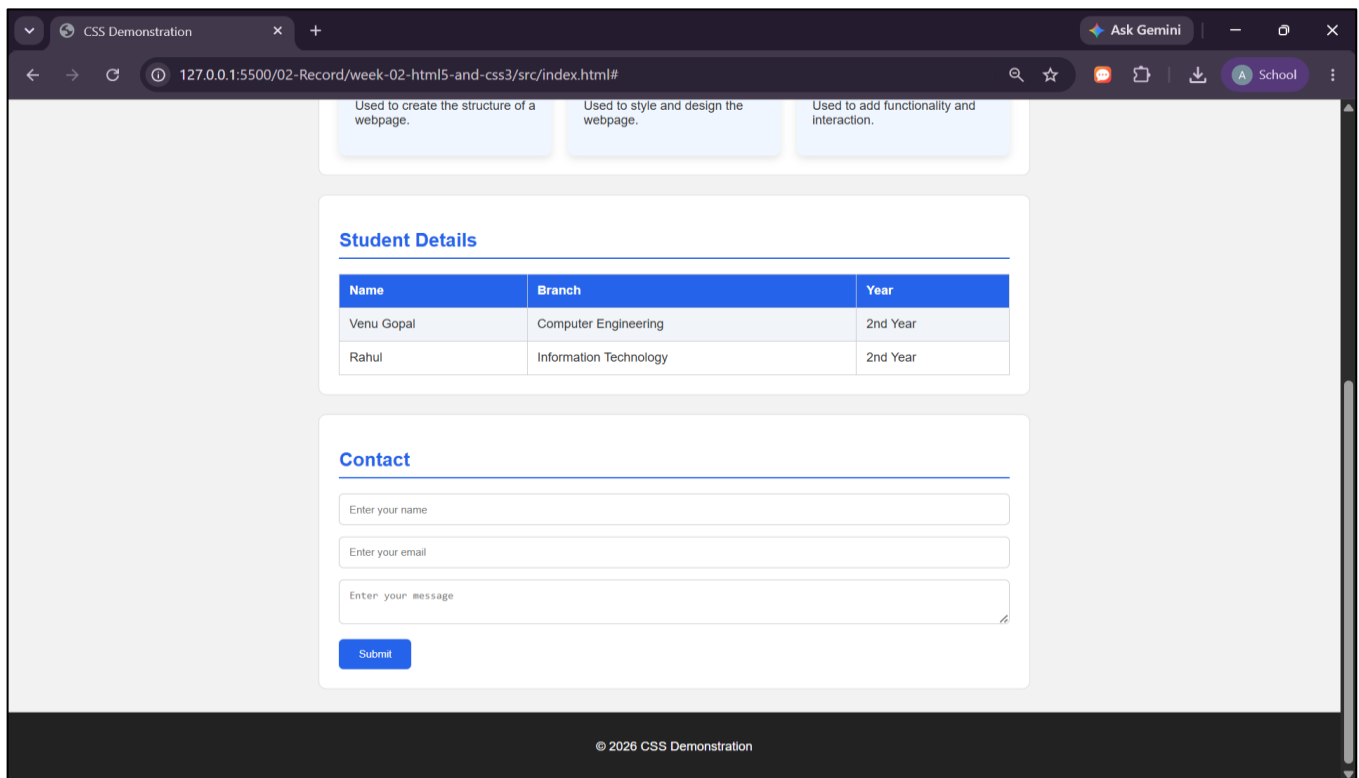
Paste the screenshots of the executed webpage below.

Screenshot 1: Main Webpage



Screenshot 2: CSS Effects / Form





8. Result / Conclusion

The webpage was successfully designed and implemented using HTML5 and CSS3. Basic and medium-level CSS concepts such as selectors, colors, box model, Flexbox, pseudo-classes, transitions, transforms, shadows and form styling were demonstrated successfully.